Samsung Nguyen

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EDUCATION

Iowa State University Expected Graduation: May 2029

Bachelor of Science, Mechanical Engineering, 4.00 GPA - President's List 2x *Ames, Iowa*

WORK EXPERIENCE

Thingz May 2026 – Aug. 2026

Mechanical Engineering Intern *Galena, Illinois*

- Designed and managed large electrical substations in Autodesk Inventor by aligning and interpreting 30M+ point LiDAR scans to produce a digital twin representative of real-world equipment and environment
- Reduced design turnaround time by roughly 50% by rapidly building and organizing a standardized library of substation parts and assemblies, streamlining the CAD process and providing reference models for future projects
- Developed accurate, detailed models of electrical substation equipment, producing 1,000+ parts and sub-assemblies
- Operated 360° LiDAR scanners and cameras on-site to capture point cloud data across an entire 5+ acre substation
- Worked cross-functionally to coordinate modeling efforts and deliver weekly verbal and written progress reports
- Identified a productivity gap and resolved it by creating assembly identifier flow charts, diagrams, and documentation
- Consistently adhered to modeling principles and quality standards to ensure compatibility between new and existing parts, assemblies, and naming conventions

EXTRACURRICULAR

Formula SAE Electric Vehicle (FSAE EV) Sep. 2025 – Present

Suspension Subsystem Member *Iowa State University*

- Independently designed and manufactured a strain gauge calibration fixture, involving statics calculations, hands-on fabrication, and welding, culminating in significant data acquisition for the team's R&D of strain gauges
- Designed and additively manufactured jig tooling to ensure precise location of critical suspension component geometries during production, such as the A-arms, push rods, and tabs on all 4 corners of the vehicle
- Oversee and own the bell crank/rocker assembly of the latest design cycle, using design tables and proper modeling practices to validate decisions alongside improving knowledge transfer and documentation
- Worked hands-on and gained certifications on a wide variety of fabrication equipment for a diverse array of projects
- Analyzed bearing specifications to determine most efficient solution, balancing packaging constraints and cost reduction
- Precisely modeled parts using parametric modeling techniques, enabling scalability and rapid design iteration to verify packaging concerns for Blade Holder and bearing integration

SolidWorks Club Aug. 2025 – Present

Sophomore Representative *Iowa State University*

- Achieved SolidWorks CSWP Certification with a flawless 100% through weekly CAD assignments and a rigorous 3-hour final assessment covering part modeling, design iteration, and assembly skills
- Evaluated engineering drawings of models and assemblies for compliance with ASME Y14.5 standards
- Elected Sophomore Representative after first year of membership, demonstrating rapid growth and mentorship
- Provided 1-on-1 instruction to club members on core SolidWorks skills and design best practices

PROJECTS

Fishing Pole Holder Sep. 2025 – Nov. 2025

- Assessed design requirements using digital calipers and paper sketches to ensure proper product function
- Created and 3D-printed final solution using SolidWorks and Bambu Lab X1C, material choice of PETG

SKILLS AND CERTIFICATIONS

SolidWorks CSWP, Autodesk Inventor, Additive Manufacturing (FDM 3D Printing), Python, Microsoft Office (Excel, Word, PowerPoint), Metal Lathe, MIG Welding, Grinder & Sander, Horizontal & Vertical Bandsaw, Table Saw, Mill, Drill Press

INTERESTS: *Fishing, Sketching, Digital Cameras, Automotive, Astronomy, Soccer, Filmmaking, Time Management*